TASK 3: AUTOMATION SYSTEM DESIGN

Smart Home Lighting System Using PIR and LDR Sensors

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System: Automated Smart Lighting

1. Objective

Design an automated lighting system that switches lights according to human presence and ambient brightness. A PIR sensor detects motion, an LDR detects light level, a microcontroller makes the decision, and a relay controls the lamp.

2. Components

Component	Purpose
Arduino/ESP32	Reads sensors and controls output
PIR sensor	Detects motion/presence
LDR	Measures ambient brightness
Relay	Switches the lamp
LED/low-voltage lamp	Lighting output

3. Flowchart

START

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Read PIR + LDR

↓

Motion detected?

YES → Check ambient light

NO → Light OFF

↓

Room dark?

YES → Light ON

NO → Light OFF

↓

Repeat continuously

4. Working

The controller continuously reads the PIR and LDR. If no motion is detected, the light remains OFF. If motion is detected, the LDR value is checked. When the room is dark, the relay turns the light ON; when it is bright, the light remains OFF. A timer can prevent rapid switching.

5. Control Logic

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START → Read PIR and LDR → IF motion → IF dark: Light ON / ELSE: Light OFF → ELSE: Light OFF → REPEAT
```

6. Advantages

• Saves energy by avoiding unnecessary operation.
• Hands-free automatic control.
• Can be extended with Wi-Fi/Bluetooth.

• Supports manual override and scheduling.

7. Safety

For an academic prototype, use an LED or low-voltage lamp. Mains wiring should only be performed by a qualified person with proper isolation and protection.

References

1. IEEE Xplore (2024), Smart Home Automation with IoT Sensor Systems for Energy Conservation using Zigbee and WiFi. DOI: 10.1109/ICESC60852.2024.10690054.

2. IEEE Xplore (2024), Smart IoT-Based Home Automation System. DOI: 10.1109/ICAI63388.2024.10851613.

3. CodeAlpha Robotics & Automation Tasks & Instructions — Task 3.